GETACHEW HAILE

GIS & Remote Sensing Researcher | Director, ICT Directorate Oromia Agricultural Research Institute (IQQO), Oromia Region, Ethiopia

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Professional Profile

ICT and geospatial professional with over 15 years of experience at the Oromia Agricultural Research Institute, combining executive ICT leadership with applied research in GIS, Remote Sensing and digital agriculture. Leads institutional ICT strategy, enterprise networking, information systems and digital transformation, while conducting geospatial and machine-learning research on soil fertility, land suitability, water resources and climate-agriculture interactions. Proven ability to lead multidisciplinary teams, deliver digital solutions and support evidence-based decision-making through spatial analysis.

Core Competencies

ICT Leadership & Governance: ICT strategic planning; ICT governance and policy development; digital transformation; IT project management; ICT service management; change management; technology innovation.

Geospatial & Data Science: spatial analysis; satellite image processing; land use/land cover mapping and change detection; geodatabase development; cartography; environmental modeling; spatial statistics; machine learning and predictive modeling; geospatial decision support systems.

Systems Development: information system analysis and design; database design and management; research information systems; web-based platforms; workflow automation; digital service development.

Networking & Infrastructure: enterprise network administration; LAN/WAN design; wireless deployment; VPN; network security; server administration; data center management.

Professional Experience

Director, ICT Directorate, and GIS & Remote Sensing Researcher

Oromia Agricultural Research Institute (IQQO) | February 2010 – Present

ICT Leadership

- Lead and manage institutional ICT operations; develop and implement ICT strategic plans, policies, standards and procedures.
- Oversee system development projects and coordinate software and database development initiatives.
- Manage enterprise networking infrastructure, servers and information systems, ensuring secure and reliable services.
- Lead digital transformation programs and provide strategic technology guidance to management.
- Supervise ICT technical staff.

Key achievements

- Led the modernization of the institute's ICT infrastructure and services.
- Improved networking and communication systems and strengthened ICT support for research and administration.
- Enhanced information management and business processes in support of institutional digital transformation.

GIS & Remote Sensing Research (Natural Resources Research Directorate)

- Conduct GIS and Remote Sensing research; analyze satellite imagery and spatial datasets; build geospatial databases.
- Produce thematic and decision-support maps; develop geospatial research methodologies.
- Carry out environmental and agricultural assessments and support research projects through spatial analysis.
- Deliver GIS and Remote Sensing training; prepare technical reports and scientific publications.

Key contributions

- Participated in soil fertility mapping and recommendation systems.
- Supported climate-agriculture research and digital agriculture projects.

- Applied machine learning techniques in agricultural research.
- Developed GIS-based outputs for natural resource management.

Education

MSc, GeoSpatial/GIS and Remote Sensing – Addis Ababa University, July 2017 Thesis: *GIS-based approach for identification of potential rainwater harvesting sites in Arsi Zone, Central Ethiopia*

BSc, Computer Science and Information Technology – Haramaya University, July 2009 Project: *Online Car Booking and Reservation System*

Technical Skills

- **GIS & Remote Sensing:** ArcGIS Pro, ArcGIS, QGIS, Google Earth Engine, ERDAS Imagine, SNAP, ENVI
- **Programming & Analytics:** Python, R, SQL; machine learning applications
- **Databases:** PostgreSQL/PostGIS, MySQL, SQL Server; database administration
- **Networking:** TCP/IP, VPN configuration, router and switch administration, wireless networks, network security management

Selected Projects

Research & GIS: soil fertility mapping and recommendation systems; agricultural suitability mapping; climate data analysis; watershed management studies; land use/land cover mapping; spatial decision support systems; remote sensing-based agricultural monitoring; machine learning in agriculture.

ICT: institutional information systems; research information management systems; ICT infrastructure modernization; enterprise network management; digital transformation programs; database development.

Publications

1. Demonstration of pre-determined ECONAT organic liquid fertilizers on tef production in Bora District, East Shewa Zone, Oromia Region, Ethiopia.

2. Land suitability evaluation for agroforestry using geospatial techniques in Genale Sub-basin, Oromia, Ethiopia.
3. Identification and mapping of groundwater potential zones using geospatial analysis for Genale-Dawa Bale Sub-basin, Oromia, Ethiopia.
4. Selection of rainwater harvesting sites using remote sensing and GIS techniques: a case study of Dawa Sub-basin, Southern Ethiopia.
5. GIS-based approach for identification of potential rainwater harvesting sites in Arsi Zone, Central Ethiopia.
6. Land suitability evaluation for *Moringa oleifera* using GIS and AHP techniques in Weyib Sub-basin, Ethiopia.
7. The impacts of land use land cover dynamics on land management: the case of Adama Zuria District, Oromia Region, Ethiopia.
8. Assessment of the nutritional quality of durum wheat genotypes grown in the southeastern highlands and midlands of Oromia, Ethiopia.

Training Delivered

GIS fundamentals and applications; remote sensing techniques; spatial analysis using RStudio; cartography and map design; research data management; ICT systems administration; computer networking; digital skills development.

Professional Training Attended

Advanced GIS and Remote Sensing; climate data analysis; machine learning applications; ICT infrastructure management; system development methodologies; network administration and security; digital transformation strategies; research methodology.

Research Interests

Digital and precision agriculture; soil fertility mapping; climate change and agriculture; geospatial modeling; machine learning for agriculture; agroforestry systems; natural resource management; spatial decision support systems; research informatics.

Professional Membership

National Soil Digital Support Tool

Languages

Afan Oromo, Amharic, English – excellent in speaking, reading and writing.

References

Mr. Teshome Bogale – General Director, Oromia Bureau of Agriculture | Teshe_2008@yahoo.com | +251 911 388 166

Assefa Taa (PhD) – Deputy Director General, Oromia Agricultural Research Institute | taaasefa@gmail.com | +251 911 840 466

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